

Chapter 04 Homework

Part A multiple choice (select the one that is best in each case. 1 point/question)

- Which of the following statements is true ?
A) Electrolyte solutions conduct electricity because electrons are moving through the solution.
B) If you add a nonelectrolyte to an aqueous solution that already contains an electrolyte, the electrical conductivity will not change.
C) When acetone, CH_3COCH_3 , is dissolved in water, a conducting solution results.
D) When ammonium nitrate, NH_4NO_3 , dissolves in water, the solution is weakly conducting and basic in nature.
- If you have an aqueous solution that contains 1.5 moles of MgCl_2 , how many moles of ions are in the solution?
A) 1.0 B) 1.5 C) 2.0 D) 4.5 E) 3.0
- If you were to draw diagrams representing aqueous solutions of (a) NiSO_4 , (b) $\text{Ca}(\text{NO}_3)_2$, (c) Na_3PO_4 , (d) $\text{Al}_2(\text{SO}_4)_3$, how many anions would you show if each diagram contained six cations?
A) 6,12,3,9
B) 6,3,18,9
C) 6,12,2,9
D) 6,12,3,18
E) 6,12,3,9
- Given a set of species below, which can be classified as strong bases?
 $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, $\text{Al}(\text{OH})_3$, $\text{Sr}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$
A) $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$
B) $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$
C) $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$
D) $\text{Ca}(\text{OH})_2$, $\text{Al}(\text{OH})_3$, $\text{Sr}(\text{OH})_2$
E) $\text{Ca}(\text{OH})_2$, $\text{Al}(\text{OH})_3$, $\text{Ba}(\text{OH})_2$
- An aqueous solution of an unknown solute is tested with litmus paper and found to be acidic. The solution is weakly conducting compared with a solution of NaCl of the same concentration. Which of the following substances could the unknown be?
A) KOH
B) CH_3COCH_3 (acetone)
C) HNO_3
D) H_3PO_3
E) KClO_3
- An experiment showed that aqueous solution of iron(II) chloride oxidize magnesium metal. The net ionic equation is shown below
 $\text{Mg}(s) + \text{Fe}^{2+}(aq) \rightarrow \text{Mg}^{2+}(aq) + \text{Fe}(s)$
What's the reaction pattern?
A) neutralization reaction
B) displacement reaction
C) exchange reaction
D) metathesis reaction
E) precipitation reaction
- Which of the following metal will not be oxidized by $\text{Pb}(\text{NO}_3)_2$?
A) Mg B) Cu C) Al D) Co E) Ni
- Which of the following solutions is the most basic?
A) 0.600 M NaCl
B) 0.200 M LiOH
C) 0.150 M $\text{Ba}(\text{OH})_2$
D) 0.200 M glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)
E) 1.0 M HI

Chapter 04 Homework

9. Which of the following statements is true?

- A) Sulfuric acid is a monoprotic acid.
- B) HCl is a weak acid.
- C) Methanol is a base.
- D) NH_3 contains no OH^- ions, and yet its aqueous solutions are basic.
- E) HF is a strong acid.

10. Which of the following statements is true?

- A) If a substance is oxidized, there must be more oxygen in the substance.
- B) If a substance is oxidized, it must gain at least one electron and form an anion.
- C) Reduction occurs if the oxidation number of an element decreases.
- D) Oxidation and reduction must occur together in a reaction.

11. Separate samples of a solution of an unknown ionic compound are treated with dilute AgNO_3 , $\text{Pb}(\text{NO}_3)_2$, and BaCl_2 . Precipitates form in all three cases. Which of the following could be the anion of the unknown salt?

- A) Br^- B) CO_3^{2-} C) NO_3^- D) S^{2-} E) OH^-

Part B: Short Answer --Write legibly and show all work for all steps in the problem. (24 points)

12. (5 points) Write the balanced molecular and net ionic equations for each of the following neutralization reactions: (a) Aqueous acetic acid is neutralized by aqueous barium hydroxide. (b) Solid chromium(III) hydroxide reacts with nitrous acid. (c) Aqueous nitric acid and aqueous ammonia react. (d) solid cadmium sulfide reacts with an aqueous solution of sulfuric acid to form gas. (e) solid magnesium carbonate reacts with an aqueous solution of perchloric acid to form gas.

13. (4 points) What is the oxidation number of the boldfaced element in (a) SO_3 , (b) AgPF_6 , (c) HNO_3 , (d) OF_2 .

14. (4 points) A sample of 70.5 mg of potassium phosphate is added to 15.0 mL of 0.0500 M silver nitrate, resulting in the formation of a precipitate. (a) What's the reaction type? (b) Write the molecular equation for the reaction. (c) What is the limiting reactant in the reaction? (d) Calculate the percent yield, if you obtain 84.2 mg of Ag_3PO_4 .

15. (2 points) Three solutions are mixed together to form a single solution; in the final solution, there are 0.2 mol $\text{Pb}(\text{CH}_3\text{COO})_2$, 0.1 mol Na_2S , and 0.1 mol CaCl_2 present. What solid(s) will precipitate?

16. (4 points) Label each of the following substances as an acid, base, salt, or none of the above. Indicate whether the substance exists in aqueous solution entirely in molecular form, entirely as ions, or as a mixture of molecules and ions. Classify the substance as a strong electrolyte, weak electrolyte or nonelectrolyte. (a) HF, (b) acetonitrile, CH_3CN , (c) NaClO_4 , (d) $\text{Ba}(\text{OH})_2$.

17. (5 points) Using the activity series (Table 4.5), write balanced chemical equations for the following reactions. If no reaction occurs, write NR. (a) Nickel metal is added to a solution of copper(II) nitrate, (b) a solution of zinc nitrate is added to a solution of magnesium sulfate, (c) hydrochloric acid is added to gold metal, (d) chromium metal is immersed in an aqueous solution of cobalt(II) chloride, (e) hydrogen gas is bubbled through a solution of silver nitrate.

Chapter 04 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (11 points)

1-5: _____

6-10: _____

11: _____

Part B short question (24 points)

12 (5 points)

13 (4 points)

14 (4 points)

15 (2 points)

16 (4 points)

17 (5 points)